
Term Project: Rigid-Body and Orbital Dynamics

PHYS1600J — Honors Physics • Global College

Instructions. Complete **one** of the two problems below. Your submission should be a written report presenting your physical model, governing equations, the key dimensionless parameters you identify, numerical estimates, and a discussion of the assumptions and limitations of your model.

Problem A: The Inverting Spinning Top

A common toy consists of a small spherical body with an attached stem, resembling a mushroom. When spun rapidly on a smooth horizontal surface, the top initially rotates with its stem pointing upward. After a few seconds, it spontaneously flips over and continues spinning upside-down on its stem. The inversion is driven by a subtle interplay of sliding friction, spin, and the geometry of the top, and has been a classic puzzle in rigid-body dynamics.

Your task

- Develop a physical model that captures the essential mechanics of the inversion.
 - Identify the key dimensionless parameters that determine whether the top flips.
 - Estimate the time required for the inversion to occur for a typical toy.
 - Explore how the behavior would change if the experiment were performed on the Moon (reduced gravity, negligible air resistance) or inside a rotating space habitat that provides artificial gravity.
 - Investigate the role of sliding friction: can a purely energy-conserving model explain the flip, or is frictional dissipation essential?
 - Use your model to propose how a “player” could control the flip by varying the initial spin rate and tilt, and suggest geometric modifications that would make the inversion as fast as possible.
-

Problem B: The Orbital Home Run

A baseball game is played on the surface of the Moon, where gravity is $g_M = 1.62 \text{ m/s}^2$ and there is no atmosphere. A powerful hitter strikes a ball so hard that the astronauts wonder: could a well-hit ball enter a low lunar orbit and eventually strike the batter from behind? The Moon can be treated as a smooth sphere of radius $R_M = 1737 \text{ km}$ and mass $M_M = 7.34 \times 10^{22} \text{ kg}$; its slow rotation (period 27.3 days) may be included if desired.

Your task

- Build a model to explore the feasibility of this “self-hitting” home run.
- Determine the combination of launch speed, launch angle, and altitude that would allow a baseball to complete at least one full revolution and return to the batter.
- Analyze the sensitivity of the return point to uncertainties in the initial conditions, and assess whether the precision required is within human capability.
- Consider how real-world factors — such as launching from a high mountain peak, the Moon’s rotation, or the presence of terrain features — alter the conclusions.
- Discuss what this orbital possibility would mean for the rules of baseball on the Moon: how might a fair lunar ballpark be designed, and what should the ruling be if a batted ball actually circles the Moon and comes back to the field?